UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #06

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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Due Date: Friday, April 17th TA's Mailbox

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CHE-301 Assignment-06

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April 2026

Contents

1 Problem A - Gas Mixture Saturation with Water

Problem A of Assignment Six entails a 5-component hydrocarbon mixture containing 60.847 wt% methane, 14.256 wt% ethane, 8.363 wt% propane, 8.267 wt% iso-butane and 8.267 wt% normal-butane.

I am tasked with solving for the maximum amount of water in lb/MMSCF at a pressure of P = 900 psia and T = $65°F$. I am told that both the dry and wet mixture follows the Peng-Robinson Equation of State (EOS) with zero BIPs (Binary Interaction Parameters) and that the thermal properties of pure water also are calculated from the Peng-Robinson Equation of State (EOS).

I started by converting the temperature from Fahrenheit to Celcius and the pressure from psia to bar.

$$T = 65°F \quad = FtoC(65°F) = 18.3333°C$$

$$P = 900\text{psia} \quad = PSIAtoBAR(900\text{psia}) = 62.0526 \quad \text{bar}$$

1.1 Mass-Molar Conversions

To begin I started with converting all the weight fractions into mole fractions. In order to do this, I set an initial basis so I could break the total flow into component flow.

$$m = 100 \quad kg$$

The individual component flows can now be solved for by multiplying the weight fraction against the total flow.

$$m_i = wt_i * m$$

$$m_{Methane} = 60.847 \quad kg$$

$$m_{Ethane} = 14.256 \quad kg$$

$$m_{Propane} = 8.363 \quad kg$$

$$m_{I-Butane} = 8.267 \quad kg$$

$$m_{N-Butane} = 8.267 \quad kg$$

From here I can convert these to moles by using the molar mass of each compound. These can be found through the CHE Calculator "molw" function.

$$\text{Methane} = molw1 = 16.0430\frac{kg}{kmol}$$

$$\text{Ethane} = molw2 = 30.0700\frac{kg}{kmol}$$

$$\text{Propane} = molw3 = 44.0970\frac{kg}{kmol}$$

$$\text{I-Butane} = molw4 = 58.1240\frac{kg}{kmol}$$

$$\text{N-Butane} = molw5 = 58.1240\frac{kg}{kmol}$$

Now, by dividing the mass flow rates by the molecular weight. I will have now solved for the molar flow rates.

$$n_i = \frac{m_i}{molw_i}$$

$$n_{Methane} = 3.7927 \quad kmol$$

$$n_{Ethane} = 0.4741 \quad kmol$$

$$n_{Propane} = 0.1897 \quad kmol$$

$$n_{I-Butane} = 0.1422 \quad kmol$$

$$n_{N-Butane} = 0.1422 \quad kmol$$

Adding these up gives a total molar mass of 4.7409 kmols. By dividing the component molar flow rates by this total flow rate gives the molar fraction of all the components. This will be denoted by z, not x or y as its the total molar fractional composition.

$$z_i = \frac{n_i}{n_{total}}$$

$$z_{Methane} = 0.8000$$

$$z_{Ethane} = 0.1000$$

$$z_{Propane} = 0.0400$$

$$z_{I-Butane} = 0.0300$$

$$z_{N-Butane} = 0.0300$$

Dry gas Mixture Data									
	Methane	Ethane	Propane	Isobutane	N-butane				
Weight Percent Fractions	60.8470	14.2560	8.3630	8.2670	8.2670				
molecular weights, kg/kmol	16.0430	30.0700	44.0970	58.1240	58.1240				
mass, kg	60.8470	14.2560	8.3630	8.2670	8.2670	=	100.0000	kg	
mass fractions, kg/kg	0.6085	0.1426	0.0836	0.0827	0.0827				
molar, kmol	3.7927	0.4741	0.1897	0.1422	0.1422	=	4.7409	kmol	
molar fractions, kmol/kmol	**0.8000**	**0.1000**	**0.0400**	**0.0300**	**0.0300**	1.0000			

Figure 1: Conversions for molar and mass compositions and flow rates

1.2 Calculation of Saturated Water Concentration

In order to solve for the water fraction in the saturated stream, I created a new molar fraction table such as Figure One (shown below).

Wet gas Mixture data										
	Methane	Ethane	Propane	Isobutane	N-butane	Water				
moles, kmols	3.7927	0.4741	0.1897	0.1422	0.1422	0.003623136	=	4.7446	kmol/s	
molar fractions, kmol/kmol	0.79938596	0.099923397	0.03997201	0.0299775	0.029977497	0.000763638	=	1.0000		
molecular weights, kg/kmol	16.0430	30.0700	44.0970	58.1240	58.1240	18.015				
mass, kg	60.847	14.256	8.363	8.267	8.267	0.065270801				

Figure 2: Conversions for wet gas mixture

I started with an initial basis of 0.01 as the total moles for water in the wet mixture. Using this will allow me to frame future equations without any unwanted errors. I used the table to solve for temporary molar fractions of each component based on a changing water quantity.

From here I used the VLE Calculations sheet in the CHE Calculator. The first step was to set up the vapor and liquid fractions based off the data. All total z_i compositions were ported from the dry mixture table (Figure One).

From the relation of dry gas to water droplets, I know that the vapor fractions are equal to the Z fractions for the hydrocarbons and the water. For the liquid compositions it is simpler as water has $x_{H_2O} = 1$, while all others are zero $x_{i-n} = 0$.

With these compositions now I solved for the liquid and vapor compressibility factors, using the CHE Calculator function "=ZComp(molar fractions, Temperature, Pressure, Z guess)".

$$Z_{Vapor} = ZComp(y_i, Temp, Pres, 1) = 0.7492$$

$$Z_{Liquid} = ZComp(x_i, Temp, Pres, 0) = 0.0541$$

Since the compressibility of the liquid is less than the compressibility of the vapor, I am able to continue solving.

The next step is to solve for the liquid and vapor fugacity for each of the components. this is done using the "PhiC(molar composition, component, temperature, pressure, Z)".

$$\hat{\phi}_{L,i} = PhiC(x_i, compID, Temp, Pres, Zl)$$

$$\hat{\phi}_{V,i} = PhiC(y_i, compID, Temp, Pres, Zv)$$

Using these equations yielded value of

$$\hat{\phi}_{L,Methane} = 23,644.5513 \qquad \hat{\phi}_{V,methane} = 0.8873$$

$$\hat{\phi}_{L,Ethane} = 22,866.6550 \qquad \hat{\phi}_{V,Ethane} = 0.5867$$

$$\hat{\phi}_{L,Propane} = 239,197.6629 \qquad \hat{\phi}_{V,Propane} = 0.4166$$

$$\hat{\phi}_{L,I-Butane} = 8,967,504.9240 \qquad \hat{\phi}_{V,I-Butane} = 0.3174$$

$$\hat{\phi}_{L,N-Butane} = 2,883,972.3429 \qquad \hat{\phi}_{V,N-Butane} = 0.2969$$

$$\hat{\phi}_{L,Water} = 0.0003 \qquad \hat{\phi}_{V,N-Water} = 0.3886$$

This then led me to solve for the k-values for each of the components where k is defined below.

$$k_i = \frac{\hat{\phi}_L}{\hat{\phi}_V}$$

$$k_{Methane} = 26647.2261$$

$$k_{Ethane} = 38973.3478$$

$$k_{Propane} = 574197.5226$$

$$k_{I-Butane} = 28255423.0729$$

$$k_{N-Butane} = 9713548.8843$$

$$k_{Water} = 0.0008$$

From here, I set a = 1 and used Goal Seek to set the Rachford-Rice Equation to zero by changing the total water amount in kmols. This returned a new value of

$$n_{water} = 0.00363002 \quad \text{kmols } H_2O$$

From here I need to convert the kmols of water to lbs of water, where the molecular weight of water and the kg to lb conversion are found from the CHE Calculator.

$$n_{water} = 0.00363002 \quad \text{kmols } H_2O * MW_{water} * \frac{2.204623 \text{ lb}}{1 \text{ kg}} = 0.144170899 \text{lbs of Water}$$

The conversions for the water are shown below.

Dry Water Conversions			
	Water		
mass	0.00363002	kmols	
molecular weight	18.0150	kg/kmol	
mass	0.065394809	kg	
unit	2.204623	lb:kg	
Mass	0.144170899	lb	

Figure 3: Water Conversions from kmols to lbs

1.3 Unit Conversion to MMSCF

At this point I know the kmols of water and kmols of dry gas mixture,

$$n_{water} = 0.00363002 \quad \text{kmols } H_2O$$

$$n_{drygas} = 4.7409 \quad \text{kmols dry gas}$$

Using these values I solve fro the mole fraction of water on a dry gas basis,

$$\frac{n_{water}}{n_{drygas}} = \frac{0.00363002}{4.7409} = 0.000765674\frac{\text{kmols Water}}{\text{kmols Dry Gas}} = 0.000765674\frac{\text{lbmol Water}}{\text{lbmol Dry Gas}}$$

Now with the molecular weight of water being known as $MW_{water} = 18.015\frac{kg}{kmol} = 18.015\frac{lb}{lbmol}$,

$$0.000765674\frac{\text{lbmol Water}}{\text{lbmol Dry Gas}} * MW_{water} = 0.013793611\frac{\text{lb water}}{\text{lbmol dry gas}}$$

Now, with the following conversions I can turn this expression into lb water per MMSCF of dry gas.

$$1 \text{ lbmol of gas} = 379.5 \text{ scf}$$

$$1 \text{ MMSCF} = 1{,}000{,}000 \text{ scf}$$

$$\therefore \qquad 0.013793611\frac{\text{lb water}}{\text{lbmol dry gas}} * \frac{1 \text{ lbmol of gas}}{379.5 \text{ scf}} * \frac{1{,}000{,}000 \text{ scf}}{1 \text{ MMSCF}} = 36.34680097\frac{lb}{MMSCF}$$

This gives the final answer of,

$$\text{Amount of Water} = 36.34680097\frac{lb}{MMSCF}$$

1.4 Solving for T_{Dew} in Fahrenheit

From here, I can now start to solve for the dew point temperature of the mixture under the given temperature and pressure. I started by converting the temperature from Fahrenheit to Celcius and the pressure from psia to bar.

$$T = 65°F \quad = FtoC(65°F) = 18.3333°C$$

$$P = 900\text{psia} \quad = PSIAtoBAR(900\text{psia}) = 62.0526\text{bar}$$

Then I moved on to developing an initial guess for the Dew point temperature. This was done using the following equation,

$$T_{Dew} = 1.1 * \sum Z_i * T_{c,i}$$

where Z_i is the total molar composition and $T_{c,i}$ is the component critical temperature. The composition was solved in the previous section, but the critical temperatures were found through the data base section of the CHE Calculator. Which returns,

$$T_{dew,guess} = -27.7000°C$$

After this, I set the $y_i = Z_i$ and $x_i = 0$. Except for isobutane and normal butane as they have the heaviest molecular weights so those were set to $x_i = 0.5$.

With these compositions now I solved for the liquid and vapor compressibility factors, using the CHE Calculator function "=ZComp(molar fractions, Temperature, Pressure, Z guess)".

$$Z_{Vapor} = ZComp(y_i, DewT, Pres, 1) = 0.4395$$

$$Z_{Liquid} = ZComp(x_i, DewT, Pres, 0) = 0.2657$$

Since the compressibility of the liquid is less than the compressibility of the vapor, I am able to continue solving.

The next step is to solve for the liquid and vapor fugacity for each of the components. This is done using the "PhiC(molar composition, component, temperature, pressure, Z)".

$$\hat{\phi}_{L,i} = PhiC(x_i, compi, DewT, Pres, Zl)$$

$$\hat{\phi}_{V,i} = PhiC(y_i, compi, DewT, Pres, Zv)$$

Fraction	Methane	Ethane	Propane	Isobutane	N-Butane	Water
$\hat{\phi}_L$	1.8825	0.1869	0.0349	0.0111	0.0068	0.0032
$\hat{\phi}_V$	0.8558	0.3228	0.1448	0.0772	0.0657	0.1177

Fugacity for liquid and vapor phases of components

This then led me to solve for the k-values for each of the components where k is defined below. They are ordered in the same way respectively

Fraction	Methane	Ethane	Propane	Isobutane	N-Butane	Water
k_i	2.1997	0.5790	0.2411	0.1437	0.1036	0.0268

Component K values

From here, I set a = 1 and used Goal Seek to set the Rachford-Rice Equation to zero by changing the dew point temperature. This returned a new value of

$$T_{Dew} = -35.3720°C$$

With a new T_{dew}, I have to perform a check. I set the values of x_i as defined by the below equation.

$$x_i = \frac{y_i}{K_i}$$

This then returned the following compositions

Fraction	Methane	Ethane	Propane	Isobutane	N-Butane	Water
x_0	0.000000	0.000000	0.000000	0.500000	0.500000	0.000000
x_1	0.415776	0.152366	0.118524	0.126468	0.167924	0.018942

Liquid Molar Compositions

Since these values are very far apart, I set the new values as the old compositions and repeat the goal seek. This is established with epsilon, which is chosen to be $\left[10^{-3}\right]$.

$$\left|x_i^0 - x_i^{new}\right| < \epsilon$$

These iterations will be repeated until the epsilon relation convergence test is satisfied.

Fraction	Methane	Ethane	Propane	Isobutane	N-Butane	Water
x_0	0.000000	0.000000	0.000000	0.500000	0.500000	0.000000
x_1	0.415776	0.152366	0.118524	0.126468	0.167924	0.018942
x_2	0.528061	0.147538	0.098263	0.097756	0.120140	0.008241
x_3	0.590188	0.141173	0.085784	0.081641	0.096113	0.005101
x_4	0.628952	0.135955	0.077674	0.071616	0.082052	0.003751
x_5	0.655403	0.131824	0.072021	0.064838	0.072894	0.003021
x_6	0.674620	0.128519	0.067859	0.059962	0.066472	0.002570
x_7	0.689232	0.125829	0.064665	0.056287	0.061722	0.002265
x_F	0.789461	0.102721	0.042230	0.032215	0.032518	0.000856

Liquid Molar Compositions Iterations

Temperature	Value	Units
T_0	-35.3712	$°C$
T_1	-39.0152	$°C$
T_2	-41.1656	$°C$
T_3	-42.5809	$°C$
T_4	-43.5836	$°C$
T_5	-44.3320	$°C$
T_6	-44.9140	$°C$
T_7	-45.3790	$°C$
T_F	-49.2045	$°C$

Dew Point Temperature Iterations

These iterations gave a final Dew Point Temperature (with unit conversions),

$$T_{Dew} = -49.2045°C = -56.5681°F$$

1.5 Solving for T_{Bubble} in Fahrenheit

The process for the bubble point is almost entirely the same, however there is a different formula for the initial guess. The only difference is the constant in front of the summation

$$T_{Bubble} = 0.7 * \sum Z_i T_{c,i} = -116.9545°C$$

Instead for the bubble point, I set the $x_i = Z_i$ and the y_i to [0.900000, 0.050000, 0.020000, 0.020000, 0.005000, 0.005000]. As these values keep the compressibility of the liquid below that of the vapor. Using the CHE Calculator function "=ZComp(molar fractions, Temperature, Pressure, Z guess)".

$$Z_{Vapor} = ZComp(y_i, BubbleT, Pres, 1) = 0.1954$$

$$Z_{Liquid} = ZComp(x_i, BubbleT, Pres, 0) = 0.2025$$

I also need to solve for the liquid and vapor fugacities and they are solved for in the same manner as above.

Fraction	Methane	Ethane	Propane	Isobutane	N-Butane	Water
$\hat{\phi}_L$	0.2286	0.0043	0.0002	0.0000	0.0000	0.0000
$\hat{\phi}_V$	0.2219	0.0048	0.0003	0.0000	0.0000	0.0000

Fugacity for liquid and vapor phases of components

I then solved for the new k-values which returned as,

Fraction	Methane	Ethane	Propane	Isobutane	N-Butane	Water
k_i	1.0302	0.8912	0.8265	0.8074	0.7713	0.5499

Component K values

From here, I set a = 0 and used goalseek to set the Rachford-Rice Equation to zero by changing the bubble point temperature. This returned a new value of

$$T_{Bubble} = -261.7390°C$$

With a new T_{Bubble}, I have to perform a check. I set the values of x_i as defined by the below equation.

$$y_i = x_i * K_i$$

This then returned the following compositions

Fraction	Methane	Ethane	Propane	Isobutane	N-Butane	Water
y_0	0.900000	0.050000	0.020000	0.020000	0.005000	0.005000
y_1	0.865844	0.066140	0.024970	0.024319	0.018705	0.000022

Liquid Molar Compositions

I set the new values as the old compositions and repeat the goal seek. This is established with epsilon, which is chosen to be $\left[10^{-3}\right]$.

$$\left|y_i^0 - y_i^{new}\right| < \epsilon$$

Solving this iteratively the same as in the dew point temperature gave a final Bubble Point Temperature (with unit conversions) of,

$$T_{Bubble} = -259.9083°C = -435.8350°F$$

Fraction	Methane	Ethane	Propane	Isobutane	N-Butane	Water
y_F	0.806278	0.096448	0.038403	0.029503	0.028813	0.000556

Component K values

2 Problem B - Fugacity Calculations

For Problem B, I am given a three component mixture consisting of methanol, ethanol, and normal butanol at a temperature of $T = 25°C$ and pressure $P = 20bar$. The molar composition is given as $z_i =$ [0.2, 0.3, 0.5], respectively.

The goal of this assignment is to solve for the fugacities of each component in the liquid mixture and as a pure component. With two conditions applied, the fugacities will be solved for under Ideal and Non-Ideal liquid mixtures yielding a total of 12 fugacities.

The first task given is to confirm the mixture is in the liquid phase. This will be done by solving the the dew and bubble point temperatures assuming that the mixture is ideal in one scenario and assuming it follows the Peng-Robinson EOS in another.

2.1 Part One - Phase Confirmation Calculations

2.1.1 Ideal Dew and Bubble Point Calculation

2.1.1.1 Ideal Bubble Point Temperature

In the problem statement, it is given that for the ideal mixture dew and bubble points, the Poynting factor cannot be ignored.

Here, I am going to solve for the bubble point temperature first. The first step is to develop an initial guess for the bubble point temperature,

$$T_{Bubble,Guess} = 0.7 * \sum z_i * T_{c,i}$$

$$= 0.7 * SUMPRODUCT(zi,' DataBase'!C4 : E4 + 273.15) - 273.15$$

$$T_{Bubble,Guess} = 104.0310°C$$

Next, I need to set up the formula for the Poynting factor.

$$\text{Poynting Factor}_i = exp\left[\frac{V_i^L}{RT}\left(P - P^{sat}\right)\right] \tag{1}$$

Now I need to solve for P(atm), $P_i^{sat}(atm)$, T(K), V_i^L, and R. I solved for all the component saturated pressures using the psat function.

$$P_i^{sat} = Psat(compID, T) \qquad \text{where T is the bubble temp in this case}$$

$$P^{sat}_{Methanol} = 3.7286atm$$

$$P^{sat}_{Ethanol} = 2.3325atm$$

$$P^{sat}_{N-Butanol} = 0.5836atm$$

$$R = 0.082057\frac{m^3atm}{kmolK}$$

$$P = 20bar = 19.7385atm$$

$$T = T_{Bubble} = 104.0310°C = 377.1810K$$

The liquid volume is determined from the rackett function in excel,

$$V_L = \frac{1}{\rho_L} = \frac{1}{Rackett()}$$

Plugging all this into equation One yields,

$$PF_{Methanol} = 1.0222$$

$$PF_{Ethanol} = 1.0027$$

$$PF_{N-Butanol} = 1.0025$$

At this point, I need to derive the relation between ideal mixtures and the k-values. The Poynting factor with the fugacity equation comes from the derivation below. The first equation to start with is the Gibbs differential under constant temperature.

$$dG = Vdp \tag{2}$$

From here I integrate from the saturated state to the current state,

$$\int_{G^{sat}}^{G} dG = \int_{P^{sat}}^{P} VdP$$

$$G - G^{sat} = V\left(P - P^{sat}\right) \tag{3}$$

From the Gibbs ideal relation I know that,

$$G = G^{IG} + RTlnf \tag{4}$$

and therefore,

$$G^{sat} = G^{IG} + RTlnf_v^{sat} \tag{5}$$

Now, combining equations 4 and 5 here produce,

$$G - G^{sat} = \left(G^{IG} + RTlnf\right) - \left(G^{IG} + RTlnf_v^{sat}\right)$$

$$G - G^{sat} = RTln\frac{f}{f_v^{sat}} \tag{6}$$

From here I can combine equations 3 and 6,

$$V\left(P - P^{sat}\right) = G - G^{sat} = RTln\frac{f}{f_v^{sat}}$$

$$V\left(P - P^{sat}\right) = RTln\frac{f}{f_v^{sat}}$$

$$\therefore \quad f = f_v^{sat} exp\left[\frac{V}{RT}\left(P - P^{sat}\right)\right]$$

$$f_L = f_v^{sat} exp\left[\frac{V_L}{RT}(P - P^{sat})\right] \tag{7}$$

The next step here is to recall the fugacity equilibrium relation,

$$f_V = f_L$$

where

$$f_L = f_v^{sat}(P^{sat}, T) exp\left[\frac{V_L}{RT}\left(P - P^{sat}\right)\right]$$

$$f_v^{sat} = x_i \hat{\phi}_i^{V,sat} P_i^{sat}$$

$$\therefore \quad f_L = x_i \hat{\phi}_i^{V,sat} P_i^{sat} exp\left[\frac{V_L}{RT}\left(P - P^{sat}\right)\right]$$

$$f_V = y_i \hat{\phi}_i^V P$$

Combining all these formulas obtains the following below,

$$y_i \hat{\phi}_i^V P = x_i \hat{\phi}_i^{V,sat} P_i^{sat} exp\left[\frac{V_L}{RT}(P - P^{sat})\right]$$

$$\frac{y_i}{x_i} = \frac{\hat{\phi}_i^{V,sat} P_i^{sat}}{\hat{\phi}_i^V P} exp\left[\frac{V_L}{RT}\left(P - P^{sat}\right)\right]$$

$$k_i = \frac{P_i^{sat}}{P} exp\left[\frac{V_L}{RT}\left(P - P^{sat}\right)\right]$$

Since I have already solved for the Poynting Factor of the components, I can simplify the formula to,

$$k_i = \frac{P_i^{sat}}{P} * PF_i$$

Inputting this relation into the excel file, returns the following k-values.

$$k_{Methanol} = 0.1931$$

$$k_{Ethanol} = 0.1185$$

$$k_{N-Butanol} = 0.0296$$

Now that I have my k-values, I can now solve for the bubble point temperature. I set the α vapor fraction to 0 and now solve the Rachford-Rice to zero.

This gave me a new bubble point temperature of $T_{Bubble} = 211.5729°C$. Recall that this was solved for based off a vapor molar fraction guess. From here I need to update the y_i's and rerun the goalseek. This is established with epsilon, which is chosen to be $\left[10^{-3}\right]$.

$$\left|y_i^0 - y_i^{new}\right| < \epsilon$$

These initial "new" guesses are derived from the k_i definition, $k_i = \frac{y_i}{x_i}$. So therefore the y_i can be found from $y_i = x_i * k_i$.

For this temperature, only one iteration was needed and which returned the following values

$$T_{Bubble} = 211.5729°C$$

$$x_{Methanol} = 0.366123$$

$$x_{Ethanol} = 0.391539$$

$$x_{N-Butanol} = 0.242337$$

2.1.1.2 Ideal Dew Point Temperature

Now, I need to solve for the dew point temperature, the earlier values may be used, just with an α and initial guess adjustment.

$$\alpha = 1 \qquad \text{instead of } \alpha = 0$$

$$T_{Dew,Guess} = 1.1 * \sum z_i * T_{c,i}$$

$$= 1.1 * SUMPRODUCT(zi,'DataBase'!C4 : E4 + 273.15) - 273.15$$

$$T_{Dew,Guess} = 319.5630°C$$

The main difference between the bubble and dew point temperature calculations is the Poynting Factor. Since at the dew point, the mixture is essentially entirely vapor, the Poynting factor isn't necessary. Here the k-values will be determined by the following formula,

$$k_i = \frac{P_i^{sat}}{P}$$

The saturated pressures will based off the new dew temperature and return values of,

$$P_{Methanol}^{sat} = 122.5075bar$$

$$P_{Ethanol}^{sat} = 81.3609bar$$

$$P_{N-Butanol}^{sat} = 36.8727bar$$

$$P = 20bar$$

Using these values returns new k_i's of ,

Fraction	Methanol	Ethanol	N-Butanol
k_i	6.1254	4.0680	1.8436

Component K values

From here I can Goal Seek the Rachford-rice Equation to zero by changing the dew point temperature. This yields the new temperature of,

$$T_{Dew} = 231.5007°C$$

The liquid fractions can be found by $x_i = \frac{y_i}{k_i}$

$$x_{Methanol} = 0.080388$$

$$x_{Ethanol} = 0.173707$$

$$x_{N-Butanol} = 0.745905$$

2.1.1.3 Ideal Temperature Summary

From the above sections, I have found that

$$T_{Bubble} = 211.5729°C$$

$$T_{Dew} = 231.5007°C$$

The given temperature of the mixture is $T = 25°C$.

Therefore,

$$T_{Bubble} < T < T_{Dew} \quad = FALSE$$

Since the temperature is below both the bubble and dew point temperature, then under the ideal conditions. This mixture is a subcooled liquid.

2.1.2 Peng-Robinson Dew and Bubble Point Calculation

The section will solve for the bubble and dew point temperatures where the mixture follows the Peng-Robinson EOS. This involves no Poynting Factor.

2.1.2.1 Nonideal Bubble Point Calculation

From here, I start again with the bubble point approximation. This can be taken from the earlier section as it doesn't depend on any EOS.

$$T_{Bubble,Guess} = 0.7 * \sum z_i * T_{c,i}$$
$$= 0.7 * SUMPRODUCT(zi,' DataBase'!C4 : E4 + 273.15) - 273.15$$
$$T_{Bubble,Guess} = 104.0310°C$$

For the y_i fraction, under the bubble point condition they will be equal to zero. Except for methanol, as its the lightest it will be assume that it is prominent and will receive $y_i = 1.0$. The liquid fractions, $x_i = z_i$

The next step is the compressibility factors for vapor and liquid mixtures.

$$Z_{Vapor} = ZComp(y_i, BubbleT, Pres, 1) = 0.6264$$
$$Z_{Liquid} = ZComp(x_i, BubbleT, Pres, 0) = 0.0522$$

Now I can solve for the component fugacities,

$$\hat{\phi}_{L,i} = PhiC(x_i, compID, Temp, Pres, Zl)$$
$$\hat{\phi}_{V,i} = PhiC(y_i, compID, Temp, Pres, Zv)$$

Fraction	Methanol	Ethanol	N-Butanol
$\hat{\phi}_L$	0.2174	0.1324	0.0345
$\hat{\phi}_V$	0.7432	0.6632	0.4888

Fugacity for liquid and vapor phases of components

From here the k-values are found by $k_i = \frac{\hat{\phi}_L}{\hat{\phi}_V}$

Fraction	Methanol	Ethanol	N-Butanol
k_i	0.2925	0.1997	0.0706

Component K values

Now by setting $\alpha = 0$, I Goal Seek the Rachford-Rice to zero by changing the guessed bubble point temperature and get the following value.

$$T_{Bubble} = 199.6403°C$$

The next step is to perform the same iteration check from problem A where I set the new values as the old compositions and repeat the goal seek. This is established with epsilon, which is chosen to be $\left[10^{-3}\right]$.

$$\left|x_i^0 - x_i^{new}\right| < \epsilon$$

The new y_i fractions are determined from $y_i = x_i * k_i$. This gave final values of,

Fraction	Methanol	Ethanol	N-Butanol
y_i	0.332695	0.389450	0.277855

Vapor Molar Fractions

$$T_{Bubble} = 198.9528°C$$

2.1.2.2 Nonideal Dew Point Calculation

The same dew point approximation can be used.

$$\alpha = 1 \qquad \text{instead of } \alpha = 0$$

$$T_{Dew,Guess} = 1.1 * \sum z_i * T_{c,i}$$

$$= 1.1 * SUMPRODUCT(zi,' DataBase'!C4 : E4 + 273.15) - 273.15$$

$$T_{Dew,Guess} = 319.5630°C$$

For dew point condition, the $z_i = y_i$ and x_i will equal 0. Since normal butanol is the heaviest it will get a value of $x_i = 1.0$.

Now I can solve for the compressibility factors.

$$Z_{Vapor} = ZComp(y_i, DewT, Pres, 1) = 0.8997$$

$$Z_{Liquid} = ZComp(x_i, DewT, Pres, 0) = 0.8495$$

Now I can solve for the component fugacities,

$$\hat{\phi}_{L,i} = PhiC(x_i, compID, Temp, Pres, Zl)$$

$$\hat{\phi}_{V,i} = PhiC(y_i, compID, Temp, Pres, Zv)$$

Fraction	Methanol	Ethanol	N-Butanol
$\hat{\phi}_L$	0.9756	0.9502	0.8642
$\hat{\phi}_V$	0.9557	0.9362	0.8690

Fugacity for liquid and vapor phases of components

From here the k-values are found by $k_i = \frac{\hat{\phi}_L}{\hat{\phi}_V}$

Fraction	Methanol	Ethanol	N-Butanol
k_i	1.0208	1.0149	0.9945

Component K values

Now by setting $\alpha = 1$, I goal seek the Rachford-Rice to zero by changing the guessed bubble point temperature and get the following value.

$$T_{Dew} = 214.2747°C$$

The next step is to perform the same iteration check from problem A where I set the new values as the old compositions and repeat the goal seek. This is established with epsilon, which is chosen to be $\left[10^{-3}\right]$.

$$\left|x_i^0 - x_i^{new}\right| < \epsilon$$

The new x_i fractions are determined from $x_i = \frac{y_i}{k_i}$. This gave final values of,

Fraction	Methanol	Ethanol	N-Butanol
x_i	0.102132	0.191942	0.705926

Liquid Molar Fractions

$$T_{Dew} = 214.3310°C$$

2.1.3 NonIdeal Temperature Summary

From the two above sections, i have solved for

$$T_{Bubble} = 198.9528°C$$

$$T_{Dew} = 214.3310°C$$

With the given problem temperature of the mixture as $25°C$.

Therefore,

$$T_{Bubble} < T < T_{Dew} \quad = FALSE$$

Since the temperature is below both the bubble and dew point temperature, then under the conditions. This mixture is a subcooled liquid.

2.2 Part Two - Fugacities of pure and mixture components, with and without an EOS

2.2.1 Pure and Mixture Fugacities under Ideal Conditions

In order to solve for the pure and mixture components, I need to recall equation seven,

$$f^L = f_v^{sat}(P^{sat}, T)exp\left[\frac{V_L}{RT}(P - P^{sat})\right]$$

I also recall that for an ideal liquid,

$$\boxed{\hat{f}_i^{ID} = x_i f_i^L}$$

So, I can solve for the fugacity of the component in the mixture by solving for the fugacity of the pure component. In order to solve for the pure component, I need to solve equation seven above.

One more relation i need to use is between saturated vapor fugacity and pressure.

$$\boxed{f_i^{sat(V)} = P_i^{sat} * \hat{\phi}_i^{sat}}$$

But since the mixture is ideal, $\hat{\phi}_i^{sat} \approx 1$, it is just P^{sat}.
The first step is to solve for the saturated vapor pressures of each component,

Saturated Pressure	Methanol	Ethanol	N-Butanol	Units
P_i^{sat}	0.1587	0.0729	0.0100	Bar

Component P^{sat} values

Since the pressure temperature are given in the problem, and R is constant, I just need to solve for V_L.

This can be solved for by taking the inverse of the Rackett.

$$V_L = \frac{1}{\rho_L} = \frac{1}{Rackett()}$$

Liquid Volume	Methanol	Ethanol	N-Butanol	Units
V_L	0.1587	0.0729	0.0100	$\frac{m^3}{kmol}$

Component P^{sat} values

Now I can solve for the Poynting factor specifically below,

$$\text{Poynting Factor}_i = exp\left[\frac{V_i^L}{RT}\left(P - P^{sat}\right)\right]$$

Liquid Volume	Methanol	Ethanol	N-Butanol
PF	1.0298	1.0461	1.0773

Component P^{sat} values

With these values now, the equation for pure component fugacity can be simplified to,

$$f_i^L = P_i^{sat} * PF$$

fugacity	Methanol	Ethanol	N-Butanol	Units
f^L	0.1634	0.0762	0.0108	bar

Pure Component fugacity values

With these pure component fugacity values, I can now solve for

$$\hat{f}_i = x_i f_i$$

fugacity	Methanol	Ethanol	N-Butanol	Units
$\hat{f}^L$	0.0327	0.0229	0.0054	bar

Mixture Component fugacity values

2.3 Pure and Mixture Fugacities Under Peng-Robinson EOS

In order to solve for the pure and mixture components, I again need to recall equation seven,

$$f^L = f_v^{sat}(P^{sat}, T)exp\left[\frac{V_L}{RT}(P - P^{sat})\right]$$

Where the component fugacity can be found from,

$$\hat{f}_i = y_i \hat{\phi}_i P$$

And then at equilibrium, I know $\hat{f}_i^L = \hat{f}_i^V$

$$\hat{f}_i = x_i \hat{\phi}_i^L P$$

With the following definition of the saturated fugacity,

$$f_i^{sat} = \phi_i^{v,sat} P_i^{sat}$$

Saturated Pressure	Methanol	Ethanol	N-Butanol	Units
P_i^{sat}	0.1592	0.0809	0.0142	Bar

Component P^{sat} values

The next step is to solve for the saturated fugacity coefficient, this can be solved for using the "PhiC" function at the saturated pressure. In order to solve for this coefficient, I need to solve for the compressibility factor of the mixture. This will be found with the Zcomp function.

$$Z_{Liquid} = ZComp(x_i, Temp, Pres, 0) = 0.0384$$

With my compressibility for the liquid mixture I can now solve for the fugacity coefficient.

Sat Fugacity Coeff	Methanol	Ethanol	N-Butanol
$\phi_i^{v,sat}$	12.1094	11.0074	7.5699

Component $\phi_i^{v,sat}$ values

With these values, I can now solve for the saturated fugacity,

Sat Fugacity	Methanol	Ethanol	N-Butanol	Units
$\hat{f}^{sat}$	1.9282	0.8902	0.1074	bar

Saturated fugacity values

From here, I need to solve for the Poynting factor.

$$\text{Poynting Factor}_i = exp\left[\frac{V_i^L}{RT}\left(P - P^{sat}\right)\right]$$

Poynting Factor	Methanol	Ethanol	N-Butanol
PF	1.0298	1.0461	1.0773

Poynting Factor for each Component

Now I have the saturated fugacity and the Poynting factor so I can now solve for the pure component fugacity.

$$f^L = f_v^{sat}(P^{sat}, T)exp\left[\frac{V_L}{RT}(P - P^{sat})\right]$$

Pure Fugacity	Methanol	Ethanol	N-Butanol	Units
f^L	1.9857	0.9312	0.1157	bar

Pure Component Fugacity Values

The final step is to solve for the component mixture fugacities,

$$\hat{f}_i = x_i \hat{\phi}_i^{sat} P$$

Mixture Fugacity	Methanol	Ethanol	N-Butanol	Units
$\hat{f}^L$	48.4376	66.0445	75.6988	bar

Mixture Component Fugacity Values

2.4 Fugacity Summary Data

	IDEAL SOLUTION		**NON-IDEAL SOLU**	
	Pure Component (bar)	**Component in Mixture (bar)**	**Pure Component (bar)**	**Compone**
Methanol	0.1634	0.0327	1.9857	
Ethanol	0.0762	0.0229	0.9312	
N-Butanol	0.0108	0.0054	0.1157	

Fugacity calculation summary table

3 References

1. ChatGPT was used on the development of structure and readability of this report. It was also used to help guide the thermodynamic derivations and to correct the work in this report.
2. Chapter Six of the textbook for CHE 301 - Chemical Engineering Thermodynamics.
3. Chapter Five of the textbook for CHE 301 - Chemical Engineering Thermodynamics.
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5. Chapter Three of the textbook for CHE 301 - Chemical Engineering Thermodynamics.
6. "Ideal Gas Law Constants." Tsladeche's Weblog, 17 Nov. 2010, utahchemicalengineering.wordpress.com/2010/11/17 gas-law-constants.